

Flavia Cristiana Pascal

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RESEARCH INTERESTS

Keywords: planetary interior structure modelling, planetary coupled interior-atmosphere modeling, climate modeling, exoplanetary atmospheric observation & characterization, direct imaging of exoplanets, planetary formation, stellar spectroscopy

EDUCATION

MPhil Planetary Science and Life in the Universe Oct 2025 - Jul 2026

University of Cambridge, UK

Graduated with Distinction Certificate of Excellence Awarded for Exceptional Habitability Challenge Project

Bachelor of Science in Astronomy Sep 2022 - Jul 2025

University of Groningen, NL

Cum Laude Distinction (8.9/10)

Honours College Bachelor's Programme Apr 2022 - Jun 2025

University of Groningen, NL

Honours Medal Award

Exchange Semester, School of Science Undergraduate Exchange Program Sep 2024 - Dec 2024

Hong Kong University of Science and Technology, HK

Dean's List Award, School of Science (Fall 2024-2025), for outstanding academic achievement

ACADEMIC RESEARCH EXPERIENCE

University of Cambridge

MPhil Dissertation: "Tracking the competition between hydrogen miscibility and atmospheric escape in sub-Neptunes: potential implications for the habitability of super-Earth descendants" (Supervisors: Professor Oliver Shorttle, Dr. James Rogers) Oct 2025 - Present

I will study how sub-Neptunes evolve under the combined influences of hydrogen miscibility and atmospheric escape in the post disk dispersal phase. Specifically, I will focus on how hydrogen miscibility-driven contraction delays may allow sub-Neptunes to retain thicker atmospheres than standard interior evolution models predict.

University of Groningen

Bachelor Thesis project: "Interior structure and evolution of partially molten super-Earths" (Supervisor: Dr. Tim Lichtenberg) Apr 2025 - Jun 2025

I improved Zalmoxis to model exoplanet interiors up to $50 M_{\oplus}$ and integrated it into the PROTEUS framework, enabling self-consistent atmosphere-interior simulations for planets larger than $2 M_{\oplus}$. Using this extended pipeline, I then simulated the thermochemical evolution of super-Earths ($1-7 M_{\oplus}$), analysing mass-dependent trends in crystallization timescales and volatile outgassing.

"Exoplanet interior structure modelling" (Supervisor: Dr. Tim Lichtenberg) Jan 2025 - Mar 2025

I developed an open-source Python package, Zalmoxis, to simulate the interior structure of rocky exoplanets by extending and optimizing an existing Earth-based code. The latest version of the model is available on GitHub and is actively maintained.

"Characterizing exoplanet surface composition with LIFE" (Supervisor: Dr. Tim Lichtenberg) Oct 2023 - Jun 2024

Using LIFE's observational simulation tool and the Planetary Spectrum Generator, I investigated mid-IR data availability for geologically significant rocky surfaces suitable for LIFE's direct imaging concept.

PUBLICATIONS

Tim Lichtenberg, Harrison Nicholls, Mara Attia, Laurent Soucasse, Patrick Bos, Mariana Sastre, Dan J. Bower, Robb Calder, Lorenzo Cesario, Emeline Decocq, Marijn R. van Dijk, Mohammad Farhat, Mark Hammond, Leoni Janssen, Tadahiro Kimura, Imre Kisvárdai, Yamila Miguel, Ioannis Panagiotou, **Flavia C. Pascal**, Raymond T. Pierrehumbert, Emma Postolec, Ben Riegler, Sara Seager, Oliver Shorttle, Stef Smeets, Hanno Spreeuw, Karen Stuitje, Shang-Min Tsai, Anna Grace Ulses (2026). *PROTEUS: A modular framework for simulating planetary evolution*. Submitted to JOSS, in review.

Nicholls Harrison, Oliver Shorttle, Tim Lichtenberg, **Pascal Flavia C.** (2026). *Beyond the mass-radius plane: Integrated radiative-convective and interior structure simulations of the exoplanet continuum*. Published in MNRAS.

Tim Lichtenberg, Mara Attia, Harrison Nicholls, Mariana Sastre, Dan J. Bower, Karen Stuitje, **Flavia C. Pascal**, Laurent Soucasse, Daniel Apai, Patrick Bos, Robb Calder, Lorenzo Cesario, Lisa Dang, Emeline Decocq, Marijn van Dijk, Mohammad Farhat, Kaustubh Hakim, Tadahiro Kimura, Imre Kisvárdai, Sebastiaan Krijt, Yamila Miguel, Ioannis Panagiotou, Emma Postolec, Martin Schlecker, Sara Seager, Anat Shahar, Oliver Shorttle, Paolo A. Sossi, Wim van Westrenen (2026). *Super-Earth Interiors Shrink by About 10% as They Crystallise*. Submitted to ApJ.

EMPLOYMENTS

University of Groningen

Teaching Assistant, Observational Astronomy

Apr 2025 - Jun 2025

Assisted students with data processing and analysis for group observational projects, graded computational assignments, supported tutorials.

Teaching Assistant, Introduction to Programming and Image Processing in Astronomy

Feb 2024 - Apr 2025

Teaching Assistant, From Atoms to Solids

Feb 2025 - Apr 2025

Supported tutorials, graded exams.

Teaching Assistant, Observational Astronomy

Apr 2024 - Jun 2024

Assisted students during observation nights with telescope operation, graded computational assignments, supported tutorials.

Trainee, Observational Astronomy

Feb 2024 - Apr 2024

Participated in telescope operation training sessions.

Teaching Assistant, Calculus 2 (for Physics)

Nov 2023 - Feb 2024

Graded bi-weekly homework assignments, midterm and final exams, led two weekly tutorials.

Teaching Assistant, Calculus 1 (for Physics)

Sep 2023 - Nov 2023

Graded weekly homework assignments, midterm and final exams, led two weekly tutorials.

SCHOLARSHIPS & GRANTS

“Professor Ioan Zenembisi” Scholarship Extension (£5,000)

July 2025

Awarded by an alumnus of my former high school, for outstanding academic performance during my Bachelor’s studies in the Netherlands.

J.C. Kapteyn “Sterrenkundig Studiefonds” Grant (€200)

May 2025

To support my participation at the Bad Honnef Physics School “A New Era in Exoplanet Atmosphere Observation and Characterisation”.

NL Scholarship (€2,500) & Marco Polo Grant (€981)

Jul 2024

To support my Exchange Semester at the Hong Kong University of Science and Technology.

“Professor Ioan Zenembisi” Scholarship (€10,000)

Jun 2022

Awarded by an alumnus of my former high school, for outstanding academic achievements and contributions, including representing the school in international competitions, during my high school years in Romania.

ACADEMIC TALKS

Planetary Science and Life in the Universe Reading Group, University of Cambridge Nov 2025

Co-led a discussion on the review paper “Constraining exoplanet interiors using observations of their atmospheres”, highlighting the connections between exoplanet atmospheres and their interior composition, and how observations can inform models of planetary structure and climate.

Bad Honnef Physics School “A New Era in Exoplanet Atmosphere Observation and Characterisation” Jul 2025

I presented a poster on my Bachelor Thesis project “Interior structure and evolution of partially molten super-Earths” to an international academic audience.

Honours College Faculty of Science and Engineering Closing Symposium, University of Groningen Jun 2025

I presented a poster on my research project “Exoplanet interior structure modelling” to the Honours College students and faculty.

Honours College Faculty of Science and Engineering Closing Symposium, University of Groningen & “Exo5 Mini-Symposium”, Kapteyn Astronomical Institute Jun 2024

I gave a talk and presented a poster on my research project “Characterizing exoplanet surface composition with LIFE” to the Honours College students as well as the faculty of the Kapteyn Astronomical Institute.

Physics-Astronomy-Mathematics Symposium, University of Groningen Jun 2023

I gave a talk and presented a poster for each of my two research projects (“Galactic Metrics: Unveiling NGC5005” and “Physics of Beyblades”) to the students and faculty of the Physics, Astronomy and Mathematics Programmes.

SUMMER SCHOOLS & WORKSHOPS

Bad Honnef Physics Summer School “A New Era in Exoplanet Atmosphere Observation and Characterisation” (Jul 2025, In-person)

Introduction to Astronomy Research Workshop: “Intro2Astro” (Jul 2024 - Aug 2024, Remote)

Sagan Summer Workshop “Advances in Direct Imaging: From Young Jupiters to Habitable Earths” (Jul 2024, Remote)

Sagan Summer Workshop “Characterizing Exoplanet Atmospheres: The Next Twenty Years” (Jul 2023, Remote)

“Introduction to Aerospace Engineering: Astronautics and Human Spaceflight” (edX course) (2022, Online)

“Introduction to Plasma Physics” (edX course) (2021, Online)

SELECTED COURSES

University of Cambridge

Core and Advanced Coursework: Current and upcoming courses relevant to my research interests include **Origin of Planets**, **Origin of Habitable Environments**, **Origin and Detection of Life**, **Origin and Evolution of Biospheres**, **Extrasolar Planets: Atmospheres and Interiors**, **Physics of the Earth as a Planet**.

Hong Kong University of Science and Technology

Personalized Minor Programme: In pursuit of a comprehensive and multidisciplinary semester, I followed both undergraduate courses (**Introduction to Aerospace Engineering**, **Particle Physics and the Universe**, **Machine Learning in Physics**, **The Earth as a Blue Planet**, **Music and Film**) and postgraduate courses (**Introduction to Oceanography**).

University of Groningen

Programming Experience (Bachelor of Science in Astronomy Courses): I completed various Python-based coding projects and assignments, individual or collaborative, in courses such as: **Introduction to Programming and Image Processing in Astronomy** (basic interactive astronomical data analysis), **Observational Astronomy** (basic data reduction and surface brightness profile fitting on a spiral galaxy), **Statistics for Astronomy** (statistical problems), **Numerical Methods** (common numerical methods in astronomical research, source detection and classification on a Fornax Deep Survey Field using SExtractor), **Waves and Optics** (two group research projects: analyzing

Jupiter's atmospheric spectral signatures using JIRAM's data from the Juno mission and infrared surface brightness profiling of M33 with the Spitzer Space Telescope), **Physics of Stars** (stellar spectroscopy project including qualitative comparison of stellar spectra, quantitative analysis of the spectrum of an unknown star, measuring elemental abundances), **Physics of Galaxies** (building a spectral energy distribution for a galaxy and analyzing various dependencies between associated characteristics such as colour, redshift and age), **Astrophysical Hydrodynamics** (simulation and analysis of cold cloud-hot medium interaction using the PyPLUTO hydrodynamics code).

Laboratory Experience (Bachelor of Science in Physics Courses): For more collaborative projects outside my Bachelor, I followed courses from the Bachelor of Science in Physics, such as **Physics of the Quantum Universe**, **Physics Laboratory 2** and **Physics Lab: Advanced Experiments 1**.

Interdisciplinary Extracurricular Experience: Honours College Courses: Outside my Bachelor, I became acquainted with topics of personal interest such as the math behind rasterisation, probability and strategic thinking in card games, debating, Shakespearean literature and storytelling in visual arts.

ENGLISH COMPETENCES

TOEFL iBT: 113/120 (overall score)

Dec 2023

Listening: 29/30 — Reading: 29/30 — Speaking: 26/30 — Writing: 29/30

CODING SKILLS

Languages: Python, MATLAB, SQL, LaTeX (and Overleaf), C++

Libraries: numpy, matplotlib, astropy, pandas, lifsim, scipy, photutils, sympy, pysme, fsps, hcipy, pyklip, emcee, PyPLUTO, etc.

Frameworks & Software: PROTEUS, SExtractor

Other: Experience with Unix/Linux environments, Git (GitHub, GitLab), VS Code, Python package development

EXTRACURRICULAR EXPERIENCES

Student Representative, MPhil in Planetary Science and Life in the Universe

Oct 2024 - Present

Organizing academic, social, and peer-support activities to strengthen cohort community.

Soprano, Cambridge Romanian Parish Choir

Sep 2025 - Present

I sing weekly during services and am now preparing to present our bilingual (Romanian and English) Christmas Carol Concert, performing a guitar solo and offering reflections that help the audience appreciate emotional and cultural significance of the psaltic repertoire.

Mentor, National Center of Excellence for Performance in Astronomy & Astrophysics of Romania

Jan 2025 - Present

I hold lectures and training sessions to help middle and high school students in Romania prepare for Astronomy and Astrophysics regional and national competitions.

Individual Tutor for Undergraduate Courses, University of Groningen

Sep 2023 - Feb 2025

I tutored several first and second year courses (Mechanics and Relativity, Physics Laboratory 1, Electricity and Magnetism, Quantum Mechanics 1, Quantum Mechanics 2).

Co-chair, "Practical Astronomy Crew" Committee, Sirius A Astronomy Study Association

Oct 2023 - Jun 2024

I guided first-year students at the University of Groningen in doing research on topics of interest, offering support for astronomical programming assignments or personal projects.

Secretary, "Stargazing" Committee, Sirius A Astronomy Study Association

Oct 2022 - Jun 2023

I helped organize stargazing outreach events for 20-30 people at University of Groningen.

AWARDS

As the only competitor from my region in Romania to advance to the international stages of the Olympiads of Astronomy and Astrophysics during my high school years (and one of only 5-10 students nationally selected to represent

Romania), my most noteworthy achievements include:

Bronze Medal , International Remote Olympiad of Astronomy	2021
Silver Medal , Global e-Competition on Astronomy and Astrophysics (substitute for the International Olympiad of Astronomy and Astrophysics during the COVID-19 pandemic)	2020
Bronze Medal , International Olympiad of Astronomy	2019
Bronze Medal , International Olympiad of Astronomy and Astrophysics	2019